AFIF ATHAULLAH PRANAJA

Electrical Engineer | Electrical Control & Panel Systems
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PROFESSIONAL SUMMARY

Applied Bachelor's student in Electrical Engineering (State Polytechnic of Malang) with hands-on experience in electrical/panel drafting (AutoCAD, Fusion 360), control-panel assembly (MCB, contactors, timers), and PLC programming (Schneider TwidoSuite, Omron CX-Programmer). Internship experience troubleshooting production machinery and reading technical drawings for power and control circuits, plus an undergraduate project designing the electrical control logic and wiring for an automated power-management system. Seeking Electrical position at a machinery manufacturer to apply technical drawing and electrical control skills on real production equipment.

EDUCATION

Applied Bachelor's Degree (Sarjana Terapan) in Electrical Engineering *2022 – 2026*
State Polytechnic of Malang (Politeknik Negeri Malang) | GPA: 3.72

Vocational Diploma, Computer & Software Engineering *2019 – 2022*
State Vocational School (SMKN) 4 Malang

TECHNICAL SKILLS

Technical Drawing & Drafting: AutoCAD (single/three-phase electrical design, industrial-voltage layouts), Fusion 360 (2D-to-3D drawing), reading power & control circuit schematics

Electrical & Control Panels: Power & control panel assembly (MCB, contactors, timers), PLC programming (Schneider TwidoSuite, Omron CX-Programmer), Modbus RTU

Embedded Systems: ESP32, Arduino/C++, sensor & relay integration, VS Code, Platform IO

Productivity Tools: Microsoft Office, Google Workspace, Adobe Photoshop, Canva

PROJECTS

IoT Load Management System for a 400 Wp Portable Solar Power Plant *2026*
Undergraduate Thesis | State Polytechnic of Malang

- Designed the electrical control logic and relay/contactor wiring for a smart-inverter load-management system controlling 4 load channels across 4 priority-based, SOC-tiered load-shedding levels
- Diagnosed and resolved a relay control fault caused by mismatched NC-contact logic, restoring correct load-switching behavior and cutting switching errors
- Programmed ESP32 firmware (C++/Arduino) to drive relay and sensor channels, reporting system status to a monitoring dashboard every 2 seconds with error of lossing data almost 0% while connecting network
- Developing a hysteresis-based relay control mechanism to reduce relay chattering near SOC thresholds

INTERNSHIP EXPERIENCE

Equipment Section Intern *2025*
PT Wijaya Karya Beton Tbk. — Pasuruan

- Performed daily maintenance and routine inspections on 3 units area of production equipment, including batching plants, cranes, and supporting machinery, maintaining equipment uptime
- Partnered with a team of 2 technicians per day to troubleshoot and resolve up to 2 mechanical and operational issues in the field
- Implemented Modbus RTU communication to acquire real-time data from power meters type “GAE EMG 20B” registers at real-time intervals with wireless communicating TCP/IP using Serial Device Server type HF2211 at Hoist Machine
- Learn how Integrated PLC-based systems with IoT architecture, enabling remote monitoring of Wire Cadging machines and Capasitor using PLC type Master-K120s K7M-DR40U
- Built a web dashboard visualizing real-time voltage, current, power, and status data from Crane Hoist Machine, refreshed real-time

ORGANIZATIONAL EXPERIENCE

Executive Chairperson, "PLC Competition 2024" | Power Control Division *2024 – 2025*
Electro Workshop, State Polytechnic of Malang

- Led a division of up to 121 members and organized a general competition attended by 32 include internal and external participants from Institute of Indonesia

Junior Staff, Commission IV (Cooperation & External Relations) *2023 – 2024*
Student Representative Council, State Polytechnic of Malang

- Managed UKM information media produced above 30 piece of content instagram feed and instagram story of social media content for one preiod (1,5 years) using Canva while coordinating internal and external organizational programs

ACHIEVEMENTS & AWARDS

Juara Harapan II, "Nusantara Writing Festival III" *June 2025*

2nd Honorable Mention, National Scientific Paper Competition | Paper: "MARINE SAVIOR: REMOTELY OPERATED VEHICLE (ROV) BERBASIS YOLOv8 SEBAGAI SOLUSI PENDETEKSI SAMPAH DI PERAIRAN DEMI MENJAGA KELESTARIAN EKOSISTEM BAWAH LAUT"

- Ranked 2nd Honorable Mention (Juara Harapan II) out of 54 teams/entries in a national-level scientific paper writing competition